DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #05C: NON-WEAPON EQUIPMENT & PROCEDURAL SYSTEM (PART 3 — TALISMANS, PORTABLE FORMATION CORES & PROFESSION / EXPLORATION TOOLS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap (`05A–05G`)**:
  - `05A`: Group 1 — Body Armor & Protective Attire
  - `05B`: Group 2 — Personal Jewelry & Wearables + 4 Legendary Accessories
  - **`05C` (This Document)**: Group 3 — **Wearable/Expendable Talismans (`ITEM190_TAL_001..024`), Portable Formation Cores (`ITEM190_CORE_001..012` & `REC140_ARR_*`), & Profession / Exploration / Beast Tools (`ITEM190_TOOL_001..006`, `ITEM_V234_TOOL_*`, `REC140_FRG_MITHRIL_CAULDRON_4`, `REC140_BST_BLOOD_COLLAR_3`)**
  - `05D`: Group 4 — Multi-Piece Set & Synergy Architecture
  - `05E`: Part 5 — Holistic Overview of All Non-Weapon Equipment
  - `05F`: Part 6 — Penultimate Deep Audit: Procedural Equipment Engine & Templates
  - `05G`: Part 7 — Grand Finale: Master End-to-End Audit of the Entire Equipment System

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Preserve Backward Test Contracts**: `v320_dormant_awakening_suite.js` Phase 9 (`lines 216–220`) calls `COMBAT.startFight(..., 'spar', { serious: false })` right after `ECO.armConsumableTrap(hero, 'P:BASIC_QI', 'COMBAT_START')` and asserts `hero.armedConsumables.length === 0`. Keep explicit `ECO.armConsumableTrap` firing as tested, while guarding **autonomous NPC trap-arming** so NPCs only arm traps before **real battles/expeditions (`serious: true`)**!

3. **100% Emergent Xianxia Authenticity**: Paper talismans burn charges when triggered (never repaired with iron whetstones!), alchemy cauldrons/tongs genuinely stabilize pill brewing, lightning-rod cores genuinely shield Tribulations, and siege spikes genuinely crack enemy sect barriers.

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1. [TAL-ARCH-01] Architectural Split-Brain on All 24 Talismans (ITEM190_TAL_001..024): Paper Talismans Become Permanent Indestructible Rings While charges vs maxCharges Fails!

Inspecting data/data_items.js (ITEM190_TAL_001..024, REC190_TAL_001..024) alongside js/economy.js:244, 274, 1132–1154, 1316–1339 uncovers **four interlocking structural bugs**:

1. **Field-Name Mismatch (def.charges vs def.maxCharges, economy.js:1329)**:
   - Every Talisman (ITEM190_TAL_001..024) and Formation Core (ITEM190_CORE_001..012) in data_items.js defines **"charges": 1 | 2 | 4** and "durability": 30 | 80.
   - Yet ECO.useConsumableItem (js/economy.js:1329) checks:

```
JavaScript
if (typeof def.maxCharges === 'number' && def.maxCharges > 1)
```

   Because def.maxCharges is **undefined** on all 36 items, multi-charge talismans (charges: 2 or 4) **never initialize talismanCharges (1 activation destroys all charges immediately if consumed)**!

2. **Equipment-vs-Consumable Storage Mismatch Breaks E06 (armConsumableTrap)**:
   - REC190_TAL_001..024 declare outType: 'EQUIPMENT' (slot: 'ACCESSORY'). When crafted in ECO.finishCraft (economy.js:274), they are pushed as gear instances into **W.sect.gear** (and npc.inv.gear / npc.equip.ACCESSORY)—**never** into W.sect.stock or npc.inv.mats!
   - Meanwhile, ECO.triggerArmedConsumables $\to$ ECO.useConsumableItem (economy.js:1316–1327) searches **only** npc.inv.pills, npc.inv.mats, and W.sect.stock! Thus a crafted Talisman (e.g. ITEM190_TAL_022 ยันต์กับดักเถาวัลย์ธรณี) **cannot be armed or consumed via E06 at all**!

3. **`ECO.mintGear` Overwrites `durability: 30` With `dur: 100` & Gives Infinite Permanent Charges (`economy.js:244`)**:
    - `ECO.mintGear` ignores `it.durability` (`30`) and `it.charges` (`1..2`), stamping `dur: 100, maxDur: 100` with `procedural: undefined` (`dur` never drops below `1`!).
    - **Contextual Absurdity**: A single paper talisman (`ITEM190_TAL_012` ยันต์เพลิงกัลป์ล้างผลาญ, `charges: 1`) worn in `equip.ACCESSORY` fires across **thousands of battles forever without ever burning up**, and could even be "sharpened" with a metal Whetstone!
4. **Clean Unified Fix**:
    - Copy `charges: it.charges || it.maxCharges || null` and `maxCharges: it.charges || it.maxCharges || null` onto minted talisman/core instances in `ECO.mintGear`, and check `const maxCh = def.maxCharges || def.charges` in `useConsumableItem` (also allowing `takeFromOwnerOrSect` to consume matching talisman instances from `npc.inv.gear`).
    - When a `cat === 'TALISMAN'` item equipped in `ACCESSORY` triggers its major combat proc (`EFX_BARRIER_SHIELD`, `EFX_HEAVENLY_THUNDER_SMITE`, `EFX_ELEMENTAL_ANNIHILATION`, `EFX_TERRAIN_BIND_TRAP`, `EFX_BLOOD_SACRIFICE_OVERDRIVE`), decrement `it.charges` (`1` per battle); when `it.charges <= 0`, cleanly consume the spent paper talisman (🔥 ยันต์เผาไหม้สลายเป็นเถ้าหลังสำแดงเดชครบจำนวนครั้ง) so Talisman Scribes (`TALISMAN`) have genuine, ongoing economic demand!
    - Wire autonomous pre-expedition / pre-raid trap arming (`NINPO_ELEMENTAL.EXPLOSIVE_TAG_TRAPPER`, `ARRAY_TALISMAN`, `WANDERING_DAO`)!

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2. `[TOOL-DEAD-02]` All Profession Crafting Tools (`REC140_FRG_MITHRIL_CAULDRON_4`, `ITEM190_TOOL_001..005`, `REC140_TAL_WIND_BRUSH_3`) Give `+0%` Real Crafting Success & `+0` Quality!

Look at the explicit promise in `data_items.js` on `REC140_FRG_MITHRIL_CAULDRON_4` (เตาโอสถเงินหยกปฐพี):

```
"effect": { "craftBonusPath": "ALCHEMY", "craftBonusScore": 12 }, "descTh": "อุปกรณ์ช่วยวิถีและงานหลอมโอสถ ช่วยคุมไฟปราณให้เสถียรขึ้นจริง"
```

And on `ITEM190_TOOL_001..005` (คีมหยกประคองไฟเตาหลอมโอสถ, ค้อนเหล็กไหลตีทบศาสตรา, พู่กันหยกผสมเกสรสมุนไพร, เสาหยกชำระพิษดิน):

```
"effects": [{ "effectId": "EFX_FORGING_STABILIZATION",
"params": { "craftSuccessBonus": 0.16, "qualityBonusChance":
0.20 } }]
```

Two-Layer Verification Proving `0.00%` Runtime Effect in `JS2-v2.6.0`:

1. **Dropped Metadata During Minting (`economy.js:237–247`)**:
   - On `REC140_*` recipe-only tools (`Mithril Cauldron`, `Mist Disk`, `Wind Brush`, `Blood Collar`), `ECO.mintGear` copies **only** `combatMult`, `defenseMult`, and `minStageReq`—**discarding `craftBonusPath`, `craftBonusScore`, `sectArrayRepair`, and `escortSafetyBonus`**!
   - On `ITEM190_TOOL_001..006`, `craftBonusPath` was written on `REC190_TOOL_00x.effect` instead of `ITEM190_TOOL_00x.effects[0].params`, so even `M.gearSuitability` (`martial.js:600`) sees `fx.params.craftBonusPath === undefined`!
2. **Zero Consumers in `ECO.successChance`, `ECO.finishCraft` & `ECO.harvestHerb` (`economy.js:227, 258, 356`)**:
   - `ECO.successChance(npc, r)` (`line 227`) and `ECO.finishCraft(W, npc, r, ctx)` (`line 258`) **never inspect `npc.equip.ACCESSORY`**! Equipping the Mithril Pill Cauldron, Jade Fire-Tongs, or Meteor Smithing Hammer grants **`+0%` success chance and `+0` pill/gear quality**!
   - Worse: `EFX_ECOLOGY_HERB_NURTURE`, `EFX_HERB_POLLINATION`, and `EFX_SOIL_PURIFICATION` (`economy.js:999–1003`) write `W.sect.herbYieldBonus` (only if consumed, never when equipped!), **and `W.sect.herbYieldBonus` is NEVER read anywhere in `ECO.harvestHerb` (`economy.js:356–367`)**!

Exact Wiring Fix:

- Preserve `r.effect` fields (`craftBonusPath`, `craftBonusScore`, `sectArrayRepair`, `escortSafetyBonus`) onto minted gear in `ECO.mintGear` and copy `craftBonusPath` onto `ITEM190_TOOL_001..006` `effects[0].params`.
- Create helper `ECO.equippedToolBonus(npc, domain)` reading usable `npc.equip.ACCESSORY`:
  - In `ECO.successChance(npc, r)`: add `(craftBonusScore / 100) + craftSuccessBonus` when `craftBonusPath === r.craft` (or `EFX_FORGING_STABILIZATION` matches `ALCHEMY`/`FORGING`).
  - In `ECO.finishCraft`: roll `qualityBonusChance` for `+0.08` quality!

- In `ECO.harvestHerb`: multiply yield by `(1 + herbToolBonus + (W.sect.herbYieldBonus || 0))` and boost T2/T3 rare herb upgrade probability by `qualityUpgradeChance` (`ITEM190_TOOL_004`)!

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3. `[CORE-WORLD-03]` Unwired Portable Formation Cores, Tribulation Shields, Blood-Sacrifice Lifespan Cost & Beast-Taming Tools

Inspecting the remaining specialized Talismans, Formation Cores (`ITEM190_CORE_001..012`), and Beast Tools reveals **five missing world-simulation connections**:

1. **Tribulation Lightning Shields Ignored During Heavenly Tribulation (`ITEM190_TAL_019` & `ITEM190_CORE_008` — `EFX_TRIBULATION_SHIELD`)**:
   - Both ยันต์รับสายฟ้าทัณฑ์สวรรค์ (`28%` mitigation) and แกนค่ายกลรับฟ้าทัณฑ์ทองแดงดารา (`32%` mitigation) exist specifically to protect elders crossing Heavenly Tribulation (`CULT.TRIBULATION` / `CULT.attemptBreakthrough`), **yet `cultivation.js` and `bottleneck.js` never check for `EFX_TRIBULATION_SHIELD`**!
   - **Fix**: In `CULT.breakthroughChance` / `CULT.attemptBreakthrough`, check usable `npc.equip.ACCESSORY` (or `BN.prepare` equipping one from `W.sect.gear`!) for `EFX_TRIBULATION_SHIELD`: boost tribulation survival / breakthrough chance by `+0.08..+0.12` and reduce failed-breakthrough injury severity by `tribulationMitigationRatio` (consuming 1 charge)!
2. **Blood-Sacrifice Talisman Skips Lifespan Cost (`ITEM190_TAL_024` — `EFX_BLOOD_SACRIFICE_OVERDRIVE`)**:
   - Declares `lifespanCostYears: 3, healthCost: 18, powerMultiplier: 2.35`, yet activating it deducts **`0` years of lifespan**! Deduct `c.lifeBonus -= 3; CULT.refreshLifespan(npc)` and `-18%` current HP when triggered in serious combat!
3. **Siege Array-Breaker Spikes & Sect Barrier Pillars Disconnected (`ITEM190_CORE_002`, `ITEM190_CORE_009`, `REC140_ARR_LODESTONE_BANNER_3`, `REC140_ARR_ASTRAL_CORE_5`)**:
   - Wire `EFX_FORMATION_DISRUPTION` (`ITEM190_CORE_002` หมุดเหล็กดำตัดชีพจรปราณ) into `EXPED.resolveSectSiege` Phase 2 (`PROTECTIVE_ARRAY_CLASH`: `+20%` enemy barrier breach!).
   - Wire `sectArrayRepair` (`12` / `25`) & `EFX_STRUCTURAL_IMMUNITY_FLOOR` (`ITEM190_CORE_009`) into `SECT.ELEMENTAL_BARRIER` / `LEYLINE_GRID` upkeep when worn by `arraymaster` or `guard` officers!

4. **Caravan Escort Charms & Spatial-Lock Nets (`escortSafetyBonus`, `ITEM190_TAL_005/006/020/021`, `ITEM190_CORE_004/006/011`)**:
    - Wire `escortSafetyBonus` (`REC140_TAL_SHIELD_CHARM_1` / `REC140_ARR_MIST_DISK_2`) into `SECT.CARAVAN_AUCTION` & `EXPED` hazard reduction; wire `EFX_SKY_LEVITATION_DRIVE` (`ITEM190_CORE_004`) to `-12%` expedition travel time; wire `EFX_SPATIAL_LOCK` (`ITEM190_TAL_008` / `CORE_011`) to block enemy `blink` & `flee` in `COMBAT`!
5. **Beast-Bone Flute & Blood-Pact Collar (`ITEM190_TOOL_006` & `REC140_BST_BLOOD_COLLAR_3`)**:
    - Wire `EFX_BEAST_INTIMIDATION` (`beastPowerSuppressionRatio: 0.28, maxBeastTierAffected: 5`) in `COMBAT.strike` (`-18%` incoming damage from `u.wild === true` beasts up to `maxBeastTierAffected`) and `+18%` wild-beast taming success in `wildlife.js`!

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4. Implementation & Verification Checklist for GPT / Gemini (`05C`)

- [ ] **Unify Talisman & Core Charge Lifecycle (`economy.js`, `combat.js`)**: Support `def.charges || def.maxCharges` in `mintGear` and `useConsumableItem`; let `armConsumableTrap` / `useConsumableItem` draw talismans from `npc.inv.gear` / `W.sect.gear`; expend charges on `TALISMAN` combat procs instead of acting as infinite permanent rings.
- [ ] **Wire All Profession Crafting & Herb Tools (`economy.js`, `data_items.js`)**: Preserve `r.effect` on `mintGear`, populate `craftBonusPath` on `ITEM190_TOOL_001..006`, apply `equippedToolBonus` in `ECO.successChance`, `ECO.finishCraft`, and `ECO.harvestHerb` (including reading `W.sect.herbYieldBonus`!).
- [ ] **Wire Tribulation, Siege, Logistics & Beast Tools (`cultivation.js`, `bottleneck.js`, `expedition.js`, `sect.js`, `wildlife.js`, `combat.js`)**: Connect `EFX_TRIBULATION_SHIELD`, `EFX_BLOOD_SACRIFICE_OVERDRIVE` lifespan cost, `EFX_FORMATION_DISRUPTION` siege breach, `sectArrayRepair`, `escortSafetyBonus`, `EFX_SKY_LEVITATION_DRIVE`, `EFX_SPATIAL_LOCK`, and `EFX_BEAST_INTIMIDATION`.